How Does a Coding Agent Work?

nestenius.se

MVP Microsoft Most Valuable Professional

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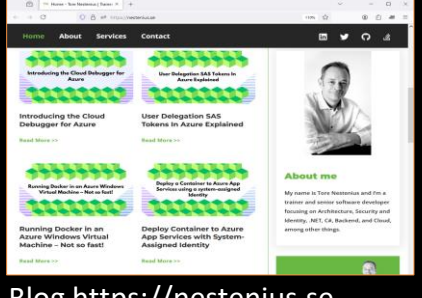
About Tore Nestenius

Work https://tn-data.se



Blog https://nestenius.se

.NET Skåne

https://meetup.com/net-skane

Have a question about this session or beyond?

tore@tn-data.se

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First, One Important Thing

Do Ask Questions!

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The AI Revolution

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The AI Revolution

Claude skapar kaos – konsultfirma slutar ta betalt för kodning: “Oundvikligt”

WIZARDWORKS SAAS AI 28 jan 2026, kl 11:26

https://www.breakit.se/artikel/45281/claude-skapar-kaos-konsultfirma-slutar-ta-betalt-for-kodning-oundvikligt

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The AI Revolution

Rebel and Bird
1,882 followers
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Är konsulttimmen död?
Vi tror det. Den traditionella modellen där man säljer tid istället för värde har nått vägs ände.
Därför genomför vi på Rebel and Bird nu vår största transformation någonsin. Vi ställer om hela verksamheten till Agentic First.
Det innebär att vi lämnar löpande räkning bakom oss och lanserar Output-As-A-Service. Genom att kombinera våra seniora konsulter med autonoma AI-agenter kan vi nu erbjuda en ekvation som tidigare var omöjlig:
-Svensk spetskompetens och fysisk närvaro.
-Korta ledtider och tät dialog.
-En prislapp som matchar offshore-leverans. Till en fast månadskostnad.

Varför välja mellan kvalitet och pris när tekniken nu tillåter båda?
Vi bygger framtiden istället för att vänta på den. Är ni redo att sluta köpa timmar och börja prenumerera på resultat?

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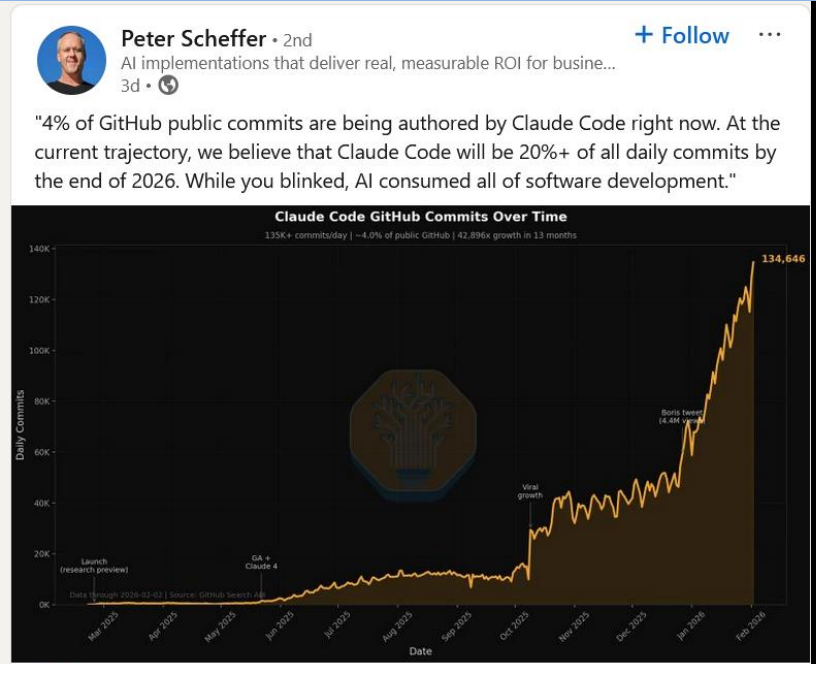
The AI Revolution



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How many of you…

Claude Code, Copilot, Cursor…

Have used a **Coding Agent**?

Could **explain** how it actually works?

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Why Did I Write This Talk?

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Why Did I Write This Talk?

I’m building an Agentic AI Development workshop

But how do you actually teach agentic dev?

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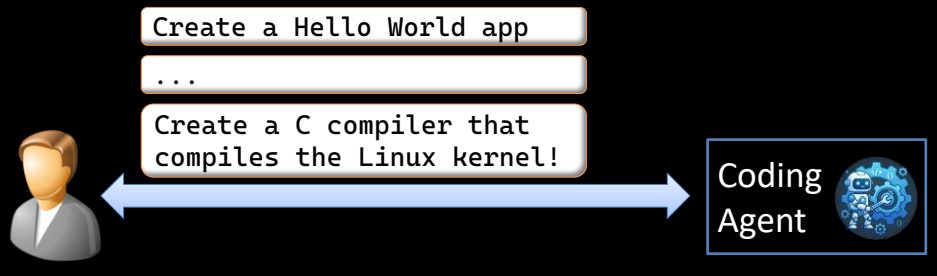
Why Did I Write This Talk?

If we don't fix this...



This will not teach you agentic coding!

What do we need instead?

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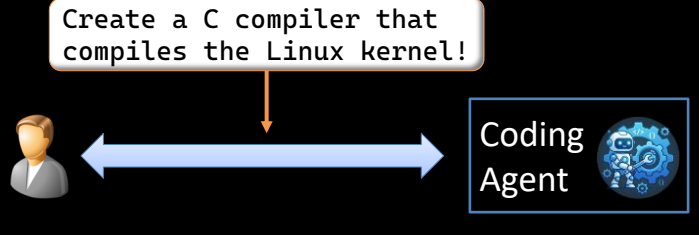
The problem with just prompting

What happens if you only learn prompting?



- Expensive
- Unreliable results
- Lack of understanding

What do we need instead?

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What do we need instead?

We need Mechanical Sympathy!

You don't drive a race car by ignoring the engine

Work with the system, not against it!

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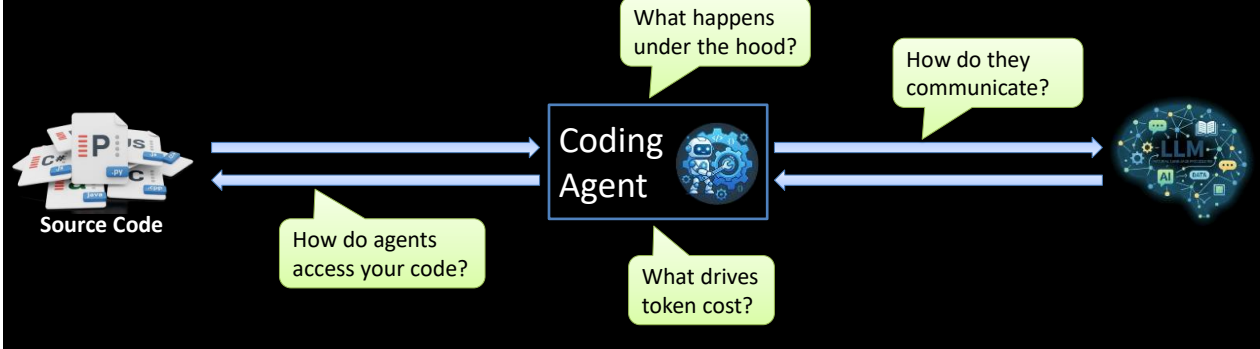
What do we need instead?

Mechanical Sympathy = understanding the system



Let’s break this down.

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From Prompting to Agentic Development

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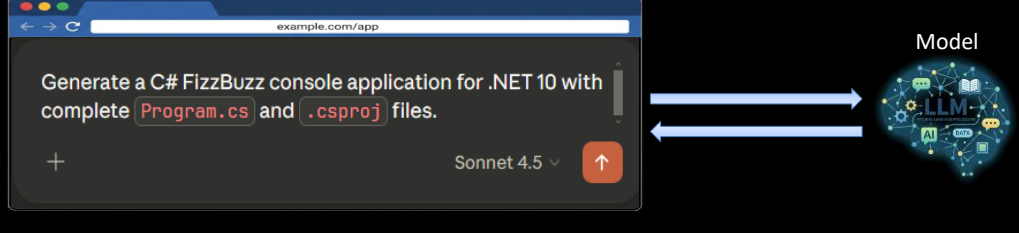
Introduction

Most of us started by using LLMs in the browser



Why doesn't this scale to real software work?

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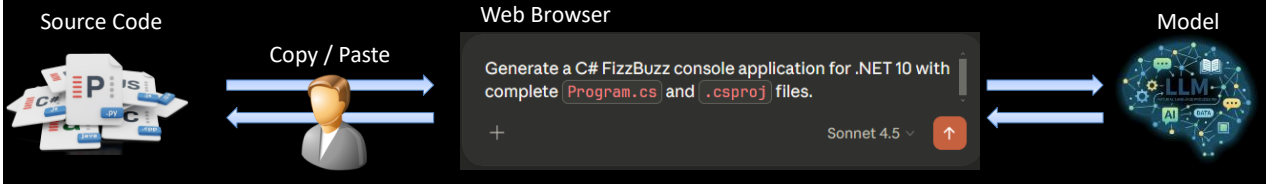
Introduction

The developer becomes the message transport



Copy -> Paste -> Error -> Copy -> Paste -> Repeat...

Can we remove the human from the feedback loop?

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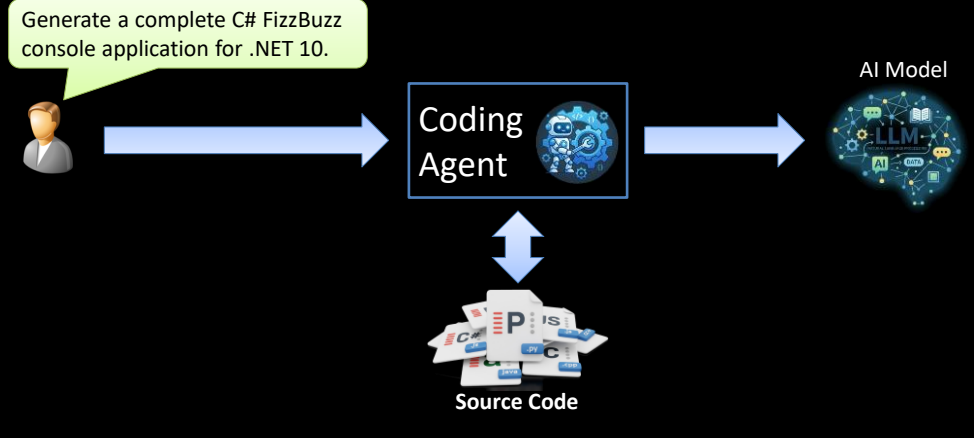
What Is Agentic Coding?

Agentic Coding: the AI operates directly on your codebase



Where do we start?

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Coding Agents

First, let's look at how the agent talks to the model

How can we observe this traffic?

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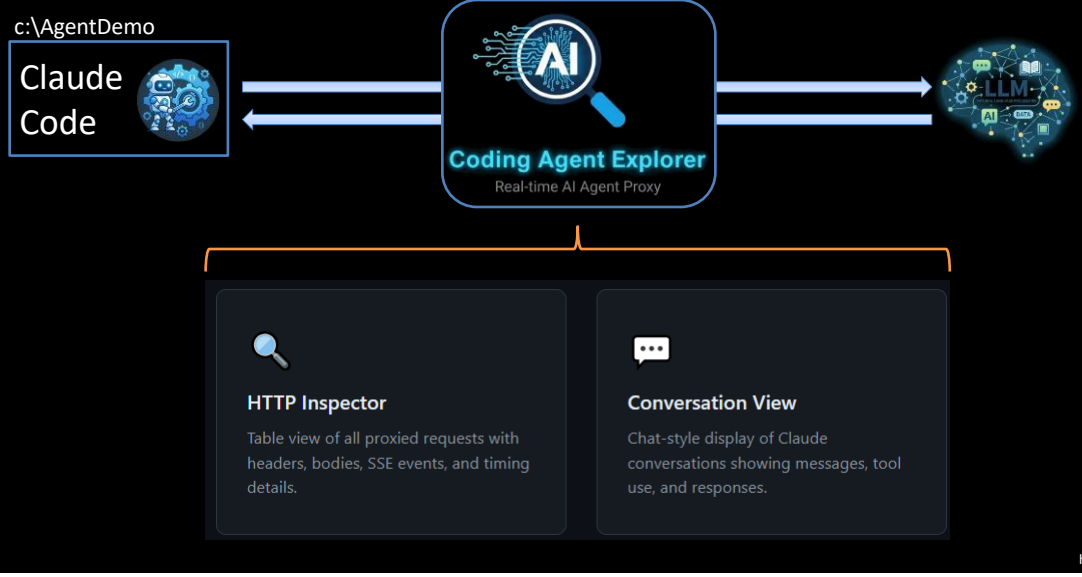
Demo #1



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How does the agent access the source code?

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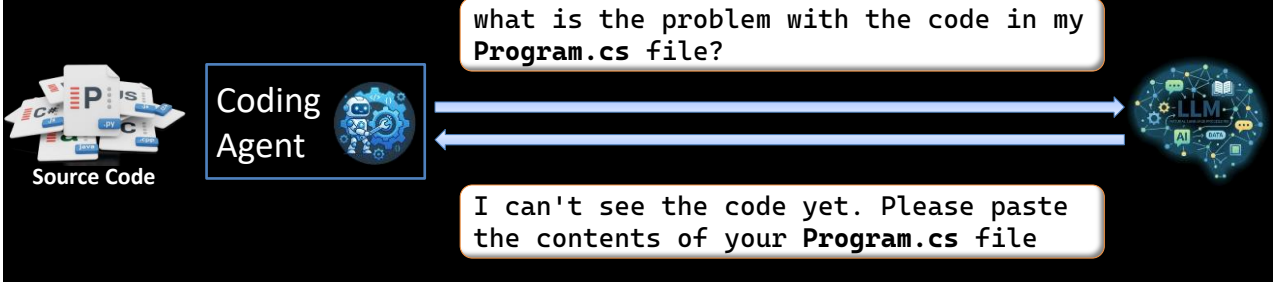
Coding Agents

Large Language Models (LLMs) operate only on text



How do we solve this?

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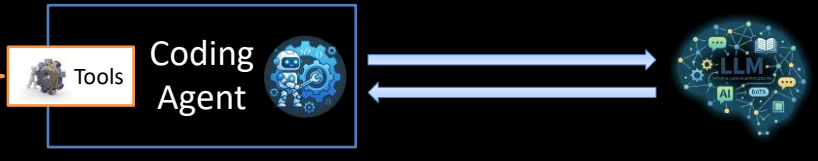
Tools

We can give the coding agent **tools**

- File I/O
- Run commands
- Web fetch
- Web search
- …



How does the model use these tools?

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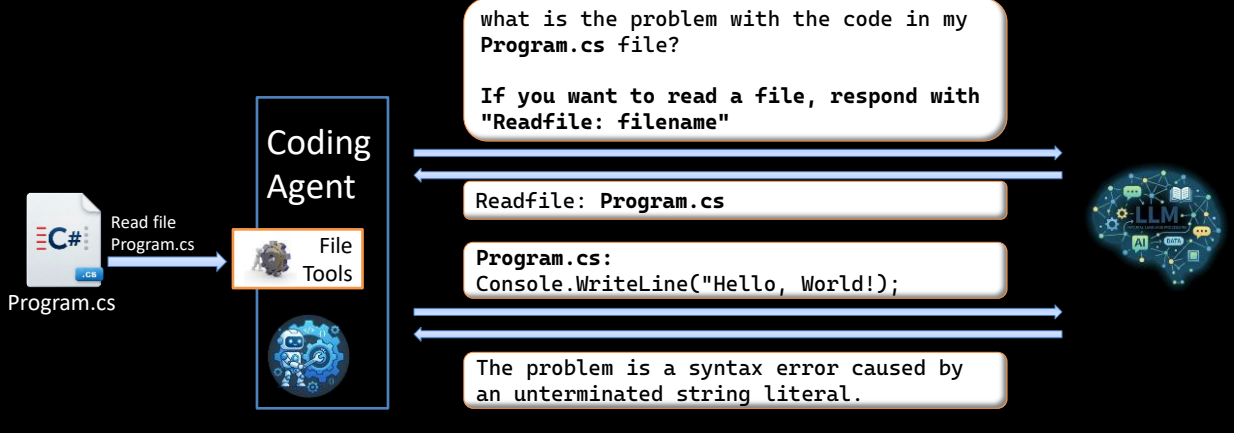
Tools

The coding agent describes the available tools to the LLM



All coding agents work in a similar way!

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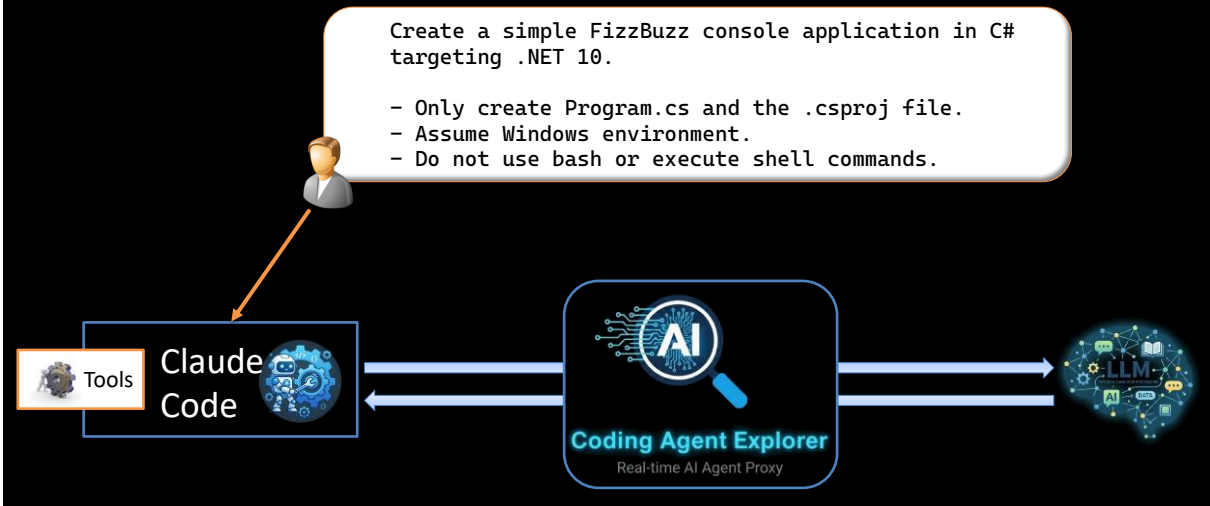
Demo #2 – Exploring the tools



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Model Context Protocol (MCP)

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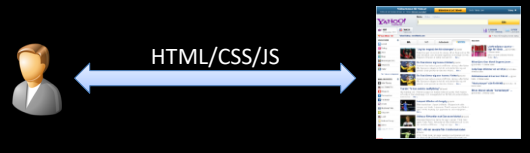
MCP

First, we had the web, for humans to consume



Then came the need for APIs

If we already have APIs, why do we need MCP?

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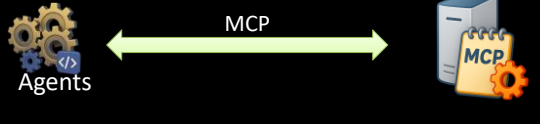
MCP

APIs are designed for applications, not agents

MCP lets you expose services and tools to agents



What does MCP specify?

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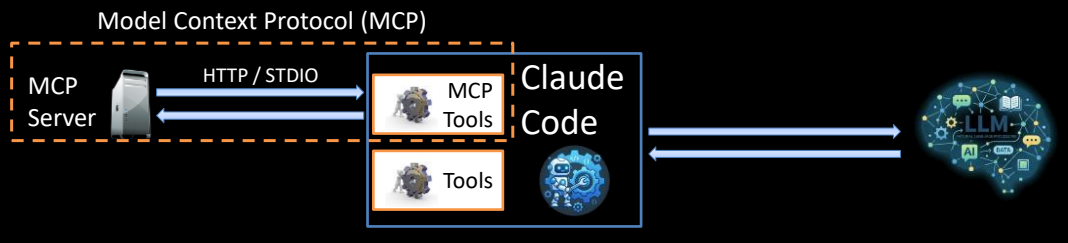
MCP Tools

MCP defines how agents communicate with external tools



How do we use these MCP tools in our coding agent?

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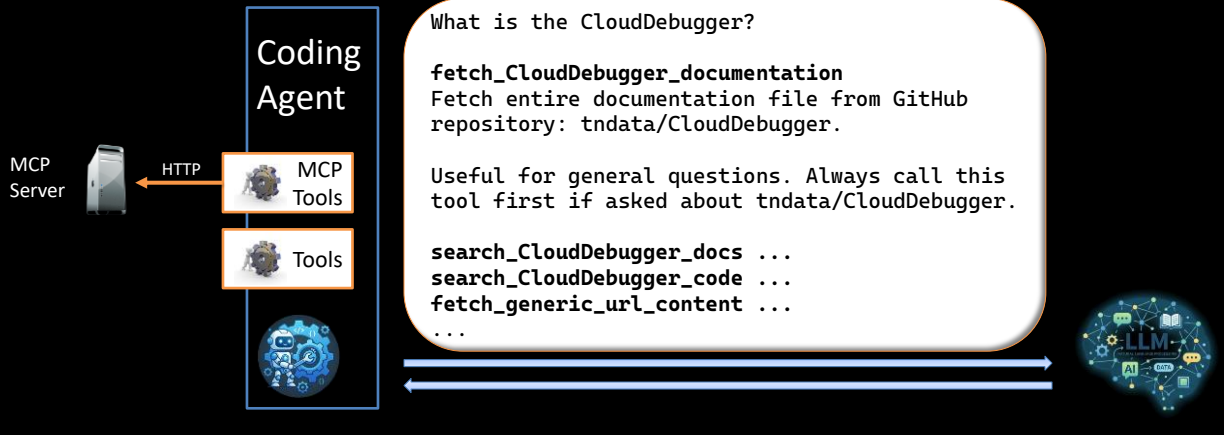
MCP Tools

MCP tools are exposed to the model as regular tools



The model never communicates directly with MCP servers

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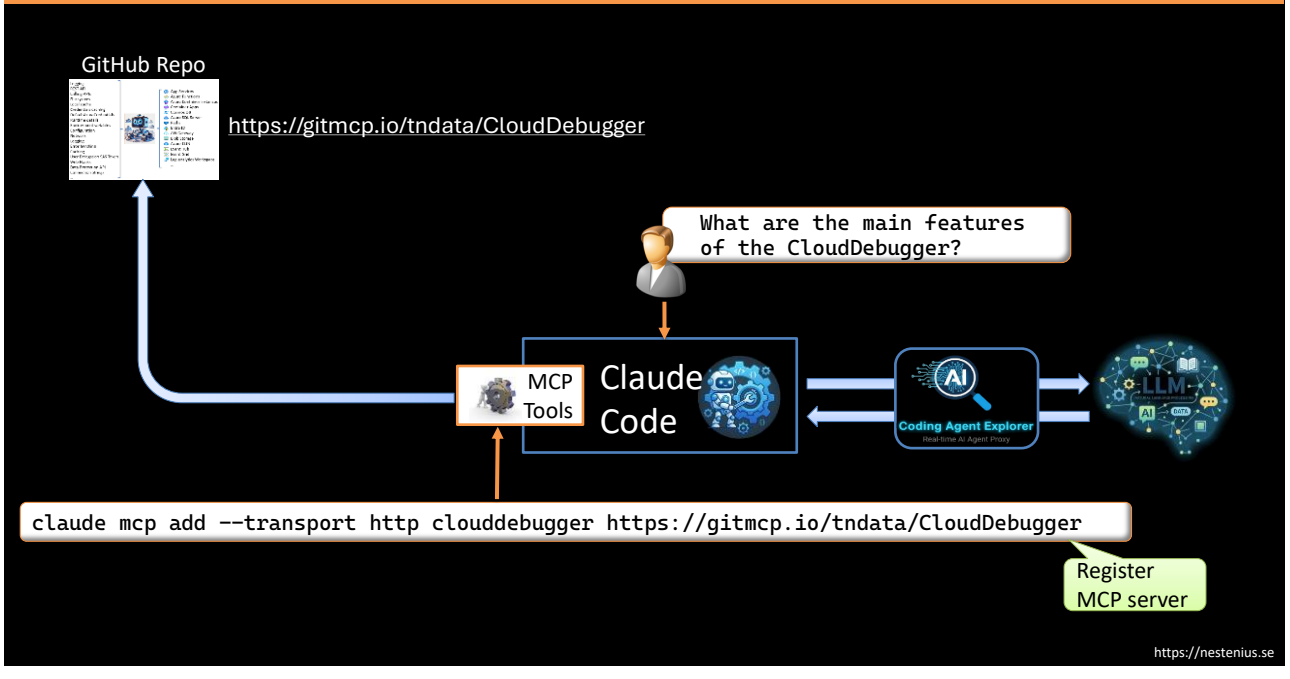
Demo #3 – Exploring MCP Usage



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Exploring the Model Context Protocol

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MCP Internals

So far, we have only seen how MCP is used by the model

How can we inspect the MCP traffic?

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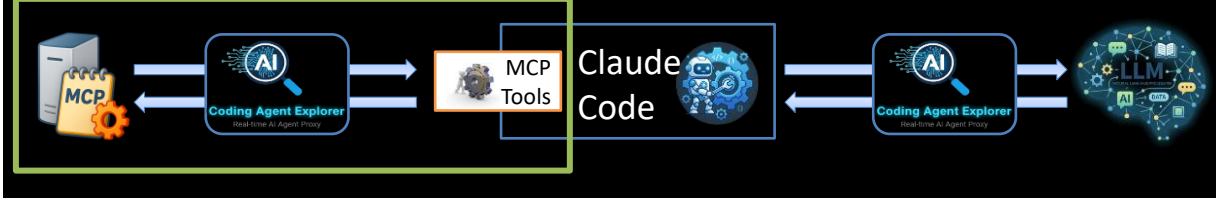
MCP Internals

We add the Explorer between the agent and MCP servers



Let's see this in action

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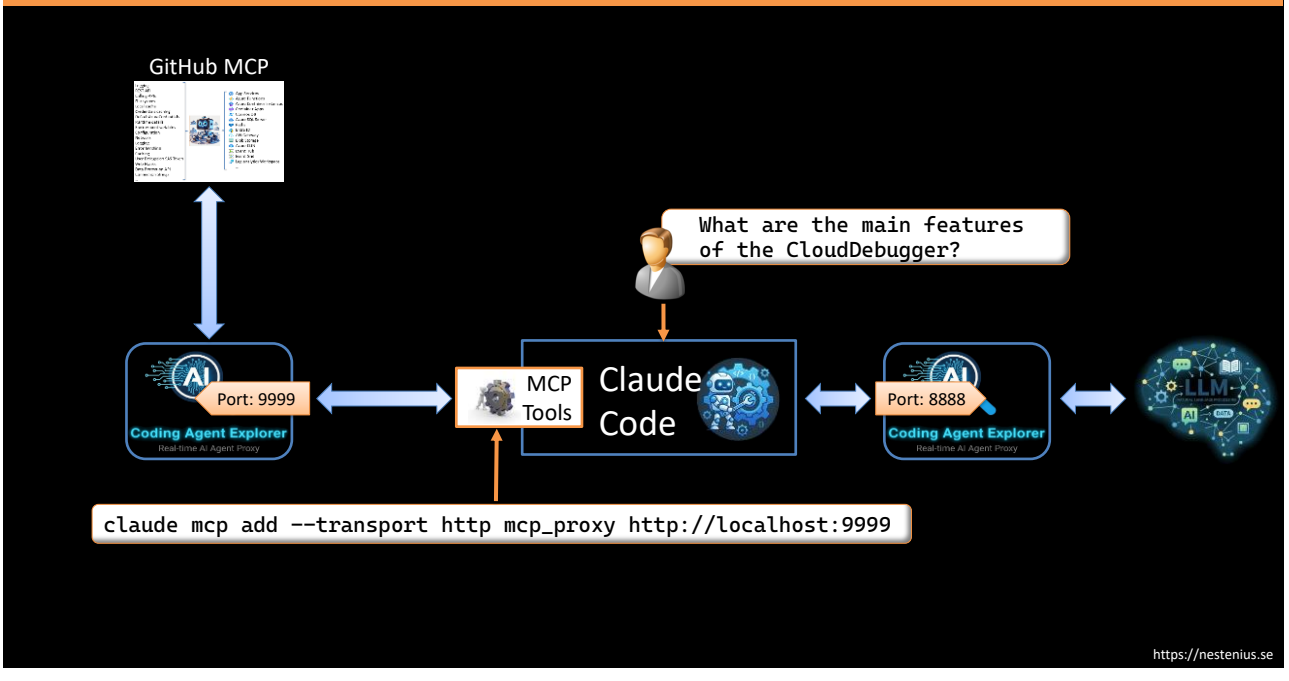
Demo #4 – Exploring MCP Traffic



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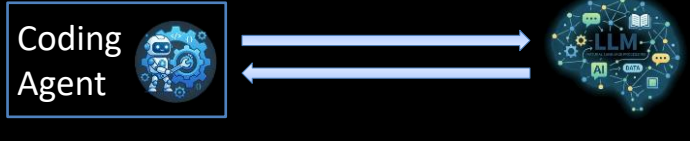
The Three Pieces of an LLM Request



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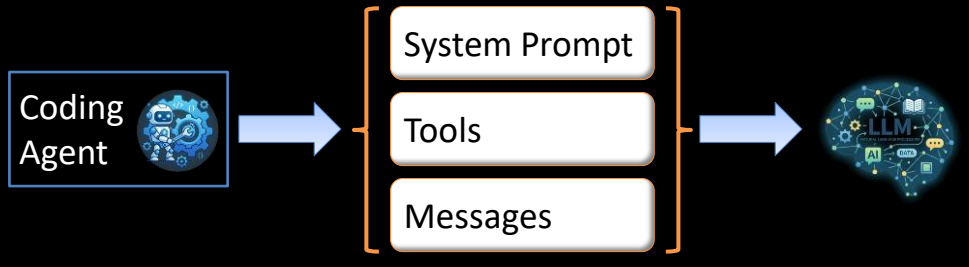
The Three Pieces of an LLM Request

Every LLM request consists of three parts:



What are the differences?

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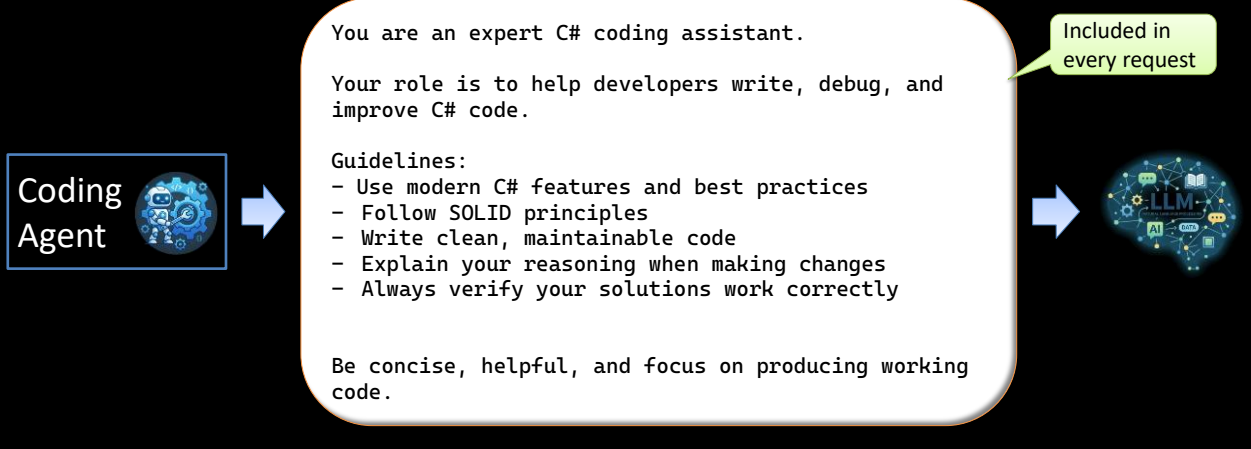
The Three Pieces of an LLM Request

System Prompt: The LLM's instruction manual



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The Three Pieces of an LLM Request

Tools: The list of available tools that the LLM can call

Coding Agent

```
Task
TaskOutput
Bash
Glob
Grep
ExitPlanMode
Read
Edit
Write
NotebookEdit
WebFetch
WebSearch
TaskStop
AskUserQuestion
Skill
EnterPlanMode
TaskCreate
TaskGet
TaskUpdate
TaskList
ToolSearch
```

```
Executes a given bash command with optional
timeout. Working directory persists between
commands; shell state (everything else) does not.
The shell environment is initialized from the
user's profile (bash or zsh).

IMPORTANT: This tool is for terminal operations
like git, npm, docker, etc. DO NOT use it for file
operations (reading, writing, editing, searching,
finding files) - use the specialized tools for
this instead.
...
```

Included in every request

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The Three Pieces of an LLM Request

Messages: The ongoing conversation

Coding Agent

```
what is the problem with the code in my
Program.cs file?

If you want to read a file, respond with
"Readfile: filename"
```

```
Readfile: Program.cs
```

```
Program.cs:
Console.WriteLine("Hello, World!);
```

```
The problem is a syntax error caused by
an unterminated string literal.
```

```
Thanks! You are a lifesaver!
```

The latest prompt

Included in every request

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Hooks

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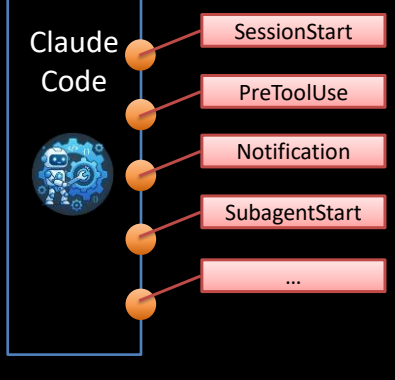
Hooks

Hooks run commands when Claude Code events occur



What can hooks be used for?

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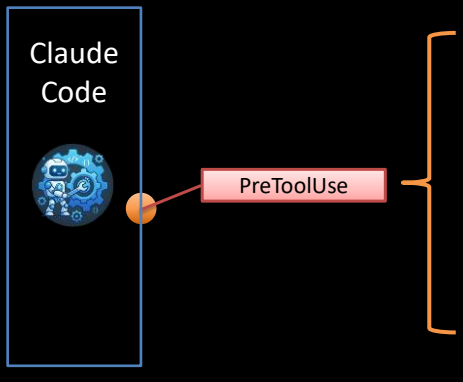
Hooks

What you can do with hooks



With hooks, you can:

- Enforce security policies before tools run
- Log and audit tool usage
- Trigger external automation
- Notify external systems (Slack, CI, etc.)

What does this look like in practice?

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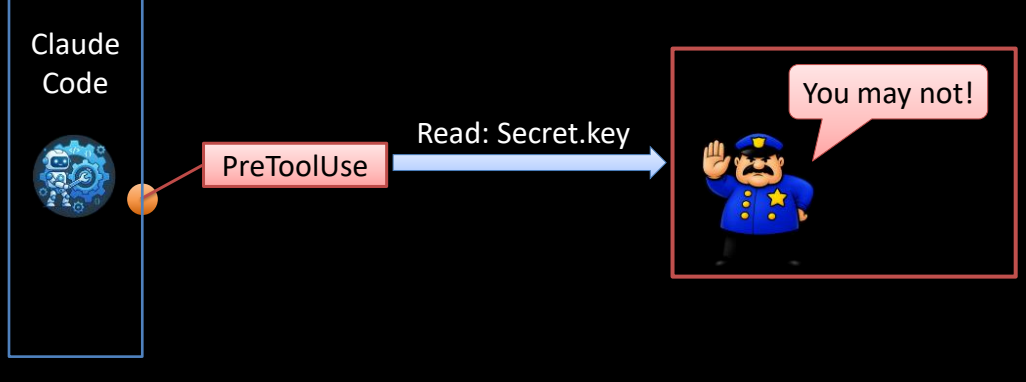
Hooks

Example: Block access to secret files



Which events can we hook into?

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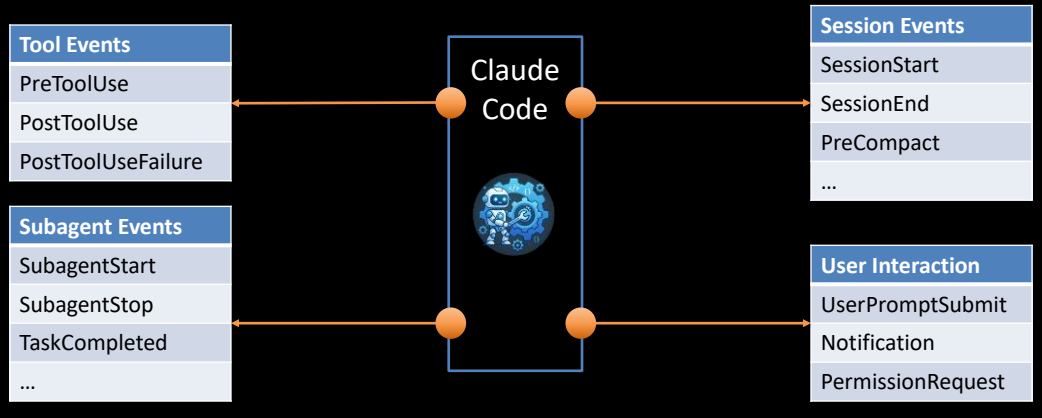
Hooks

Claude Code exposes four categories of **hookable events**



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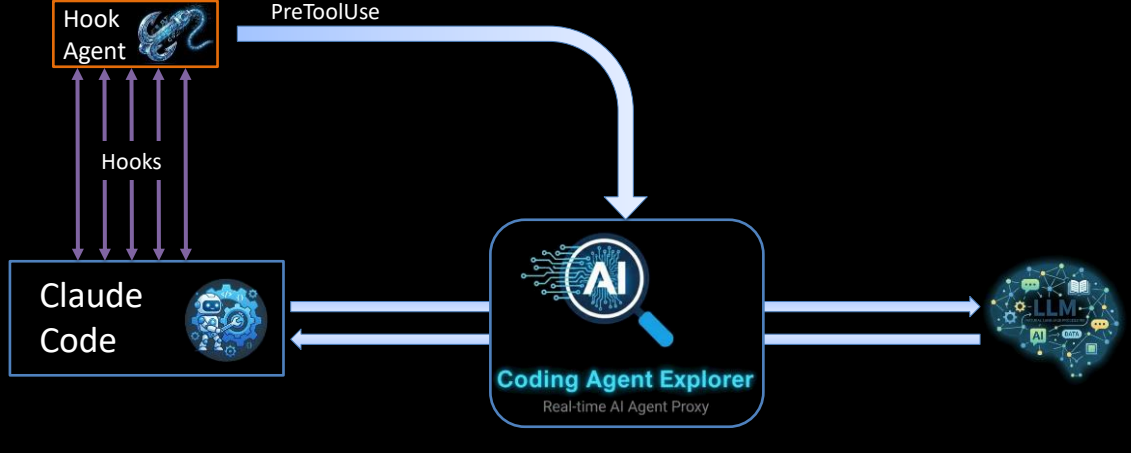
Demo #6 – Exploring Hooks



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Building Your Own Agent in C#

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Building Your Own Agent

An agent is just a loop:

```
var agent = new Agent();

while (true)
{
    var input = GetUserInput();

    var response = agent.SendMessage(input);

    Print(response);
}
```

What about the message history (context)?

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Building Your Own Agent

The context is just a collection

```
var context = new ContextCollection();

while (true)
{
    var input = GetUserInput();

    context.AddUserPrompt(input);

    var response = agent.SendMessage(context);

    context.AddAgentResponse(response);

    Print(response);
}
```

What about tool support?

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Building Your Own Agent

Adding tool support is easy:

```
while (true)
{
    var input = GetUserInput();

    context.AddUserPrompt(input);

    var response = agent.SendMessage(context);

    // Tool chain: read → edit → test
    while (agent.WantsToUseTool(response))
    {
        // Executes tool + appends tool result to context
        response = agent.ProcessToolCall(context, response);
        response = agent.SendMessage(context);
    }

    context.AddAgentResponse(response);
    Print(response);
}
```

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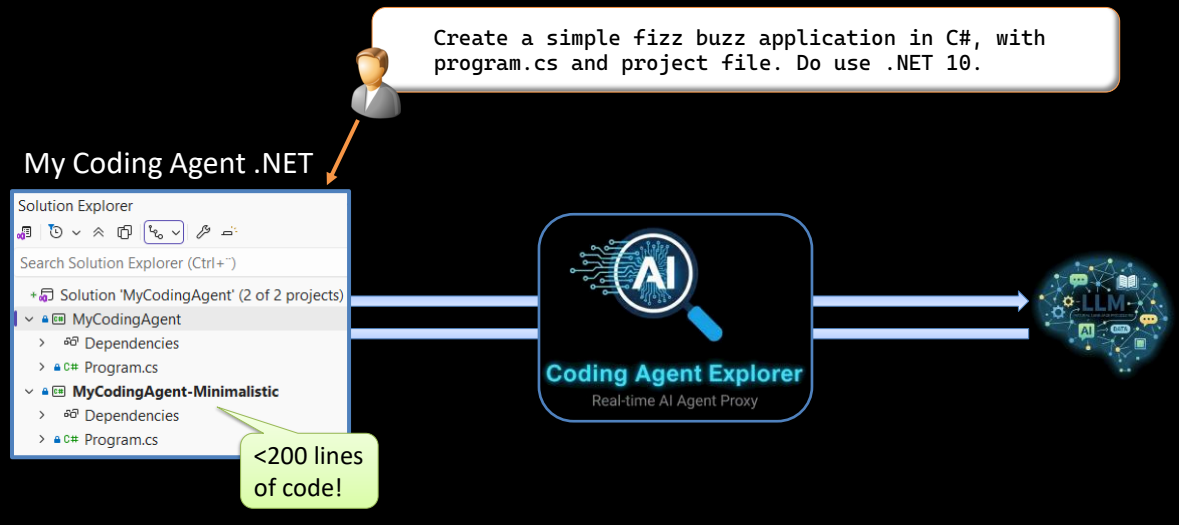
Demo #7 – MyCodingAgent.NET



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This pattern isn't just for coding!

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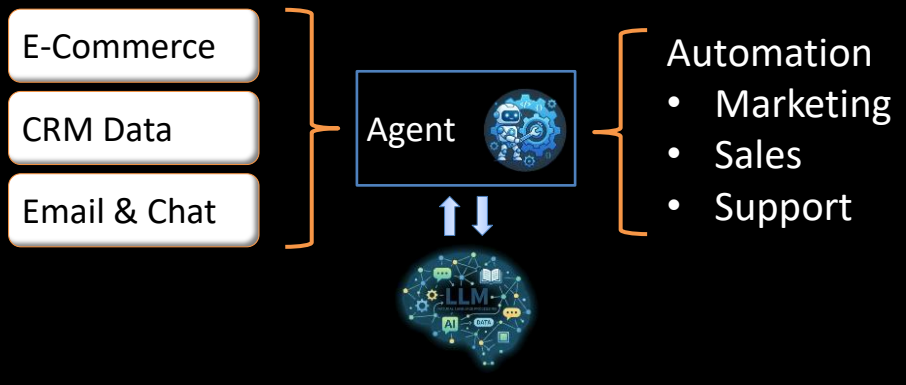
Agents Beyond Coding

The same agent loop powers automation



Agent + tools + LLM = a general automation engine

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Questions?

Your next steps:

Scan for slides + Resources

- ☑ Book this presentation for my company
- ☑ Run the Coding Agent Explorer
- ☑ Follow Tore on LinkedIn

github.com/tndataab/PublicBlogContent tn-data.se

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